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Studierichting: Informatica
Oefeningenreeks 6A nr. 10.6

$$\begin{cases} x_1 + x_1 q + x_1 q^2 = 39 \\ x_1^2 + x_1^2 q + x_1^2 q^2 = 819 \end{cases}$$

$$\Rightarrow \begin{cases} x_1(1 + q + q^2) = 39 \\ x_1^2(1 + q^2 + q^4) = 819 \end{cases}$$

$$\Rightarrow \begin{cases} x_1^2(1 + q + q^2)^2 = 1521 \\ x_1^2(1 + q^2 + q^4) = 819 \end{cases}$$

$$\Rightarrow \frac{1 + q^2 + q^4}{(1 + q + q^2)^2} = \frac{819}{1521} = \frac{7}{13}$$

$$\Rightarrow 13(1 + q^2 + q^4) = 7(1 + q + q^2)^2 = 7(q^4 + 2q^3 + 3q^2 + 2q + 1)$$

$$\Rightarrow 13(1+q^2+q^4) - 7(q^4+2q^3+3q^2+2q+1) = 6q^4 - 14q^3 - 8q^2 - 14q + 6 = 0$$

$$\Rightarrow 3q^4 - 7q^3 - 4q^2 - 7q + 3 = 0$$

$$\Rightarrow (3q - 1)(q - 3)(q^2 + q + 1)$$

$$\Rightarrow q \in \left\{3, \frac{1}{3}\right\}$$

$$x_1 = \frac{13}{1 + q + q^2}$$

Als $q = 3 \Rightarrow x_1 = 3 \Rightarrow (3, 9, 27)$

Als $q = \dfrac{1}{3} \Rightarrow x_1 = 27 \Rightarrow (27, 9, 3)$

References